

CAN

Controller Area Network

Miguel Leitão, 2015

Contents

- Introduction to CAN
- CAN Protocol
 - Physical Layer
 - Data Link Layer
- CAN Controller
- CAN Open

CAN

- CAN: Controller Area Network
- Desenvolvida pela Bosch desde 1981
- Vocacionada para a transmissão de dados entre unidades electrónicas.
- É a rede mais aplicada em veículos automóveis.
- Aplicada também na ligação de controladores industriais.

Features

- Multi-master/broadcast
- Maximum signaling rate of 1Mbps
- Absence of node addressing
 - Message ID (11 bit): contents, priority
- CSMA/CD with AMP = CSMA/BA
 - AMP = Arbitration Message Priority
 - BA= Bitwise Arbitration
- Asynchronous Serial Bus
- Simple 2-wire differential bus
- Small Messages (0-64 bits)

Non-automotive CAN Applications

- Electronically controlled production and packaging equipment
 - Machine tools; machines for molding, weaving, knitting, and sewing; systems for folding and wrapping; etc.
- Industrial freezers, printing machines
- Ships, locomotives, railway systems
- Farm and construction machinery
- Semiconductor manufacturing equipment
- Building automation: HVAC systems, elevators, etc.
- Hospital patient-monitoring systems
- Many others



Standard/Extended CAN

- CAN Versions

Name	Max Length	Max Rate	Max data	Identifier	
				Standard	Extended
CAN-LS	10 km	1 Mbps	8 Byte	11 bit	29 bit
CAN 2.0	40 m	1 Mbps	8 Byte	11 bit	29 bit
CAN-FD	5-40 m	8 Mbps	64 Byte	11 bit	29 bit

CAN-LS = CAN Low Speed

CAN-HS = CAN High Speed = CAN Classic = CAN 2.0

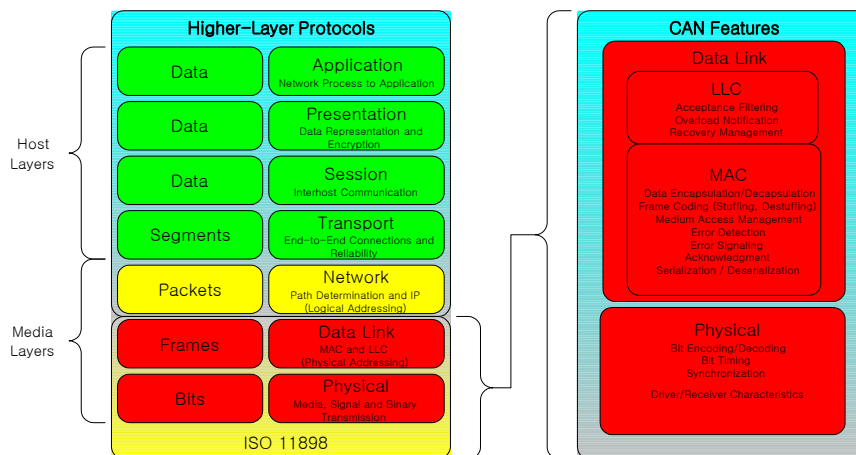
CAN-FD = CAN Flexible Data Rate

Vantagens

- Simplicidade.
- Boa imunidade ao ruído eletromagnético.
- Alta fiabilidade.
- Prioridade configurável.
- Baixo custo.
- Latência reduzida.
- Volume e peso reduzidos.
- Adotada globalmente.

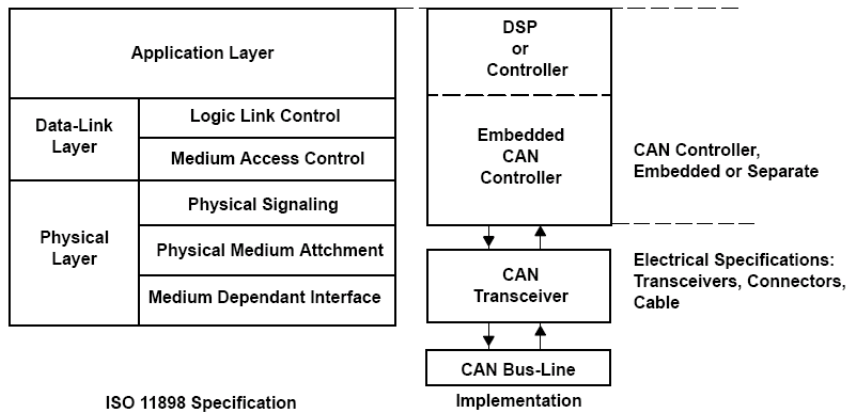
Protocol	Max. Bit Rate (Mbit/s)	Max. Data Payload (bytes)	Max. Msgs/s	MAC Efficiency	Chip Cost	Development Support
CAN	1	8	8,000	High	\$6	Substantial
LonTalk	1.25	229	560	Moderate	\$5	Substantial
IEEE-1394	400	varies, 2,048 typically	800,000	High	\$30	Moderate

CAN in the OSI Model



CAN defines only the Data Link Layer and part of the Physical Layer.

Architecture



CAN High Layer Protocols

- CAN protocol defines only the data link and part of the physical layer.
- The HLP defines :
 - Start-up behavior
 - Definition of message identifiers for the different nodes
 - Flow control
 - Transportation of messages > 8 Bytes
 - Definition of contents of Data Frames
 - Status reporting in the system

CAN High Layer Protocols

- Existing HLPs
 - CANopen
 - DeviceNet
 - CAN Kingdom
 - SAE J1939
 - X-by-wire systems & Time-triggered protocols
 - OSEK / VDX

CAN

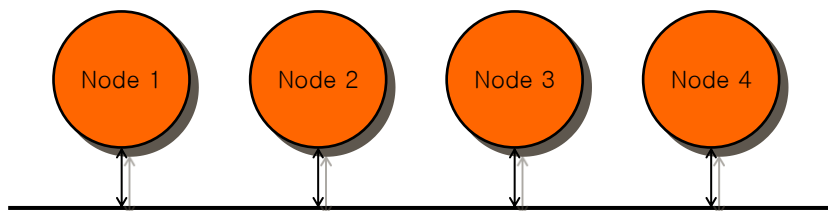
Physical Layer

CAN Physical Layer

- CAN Transceiver
 - Driver/Receiver Characteristics
- CAN Controller
 - Bit Encoding/Decoding
 - Bit Timing
 - Bus Failure
 - Synchronization Management

Topologia

- Barramento partilhado.
- Multi-Master.
- Event-triggered.
 - Qualquer nó pode iniciar uma transmissão quando desejar.



CAN Physical Layer

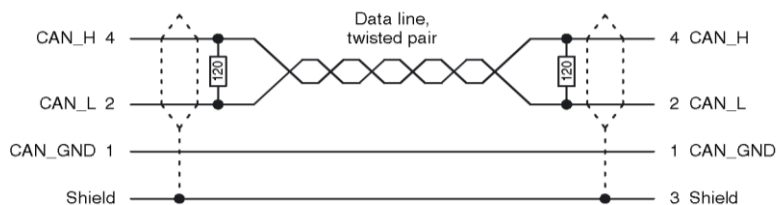
CAN Physical Layer is standardized by ISO 11898-2.

Other options are also available:

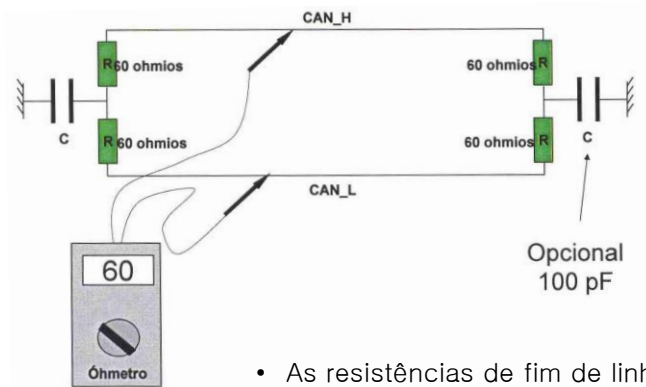
- SAE J2411 Single Wire
 - Unshielded single wire
 - Nominal data rate of 33,3 kbit/s
 - Up to 32 nodes per network.
 - Mainly use in comfort electronics networks in vehicles.
- ISO 11992-2 Point-to-Point,
 - Mainly use in vehicles with trailers.
 - 125 kbit/s with a maximum bus line length of 40 m
- ISO 11898-3 Fault-Tolerant
 - Up to 125 kbit/s
 - Up to 32 nodes

Suporte Físico

- Par de cabos entrançados :
 - Can_H
 - Can_L
- Cabos interligados por resistências nos extremos.
 - Melhora imunidade ao ruído
 - Facilita a deteção de quebras no barramento
 - Diminui reflexões nos extremos



Resistências de terminação



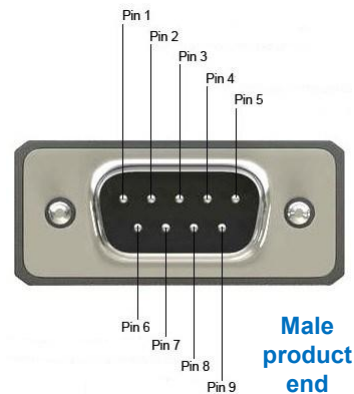
- As resistências de fim de linha podem também servir para comprovar a continuidade do barramento.
- Um ohmímetro ligado entre as linhas Can_H e Can:L marcará 60Ω.

Recommended connections

9-pin SUB D connector DIN 41652 DE-9

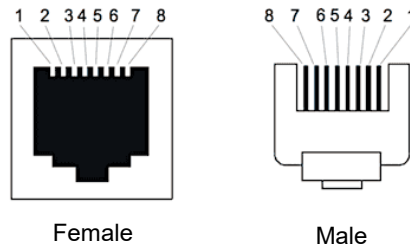
Pin Signal Description:

- 1: Reserved
- 2: CAN_L = CAN_L bus line dominant low
- 3: CAN_GND = CAN Ground
- 4: Reserved
- 5: (CAN_SHLD) Optional CAN Shield
- 6: (GND) Optional Ground
- 7: CAN_H = CAN_H bus line dominant high
- 8: Reserved
- 9: (CAN_V+) Optional CAN external positive supply



Recommended connections

RJ45 connector

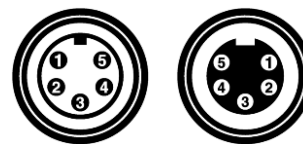


Pin Signal Description

- 1: CAN_H = CAN_H bus line (dominant high)
- 2: CAN_L = CAN_L bus line (dominant low)
- 3: CAN_GND = Ground/0 V/V-
- 4: Reserved
- 5: Reserved
- 6: (CAN_SHLD) = Optional CAN Shield
- 7: CAN_GND = Ground/0 V/V-
- 8: (CAN_V+) = Optional CAN external positive supply

Recommended connections

5-pin Mini Style connector: ANSI/B93.55M



Male

Female

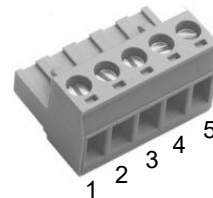
Male product end

Pin Signal Description:

- 1: (CAN_SHLD) = Optional CAN Shield
- 2: (CAN_V+) = Optional CAN external positive supply
- 3: CAN_GND = Ground/0V/V-
- 4: CAN_H = CAN_H bus line (dominant high)
- 5: CAN_L = CAN_L bus line (dominant low)

Recommended connections

Open Style connector

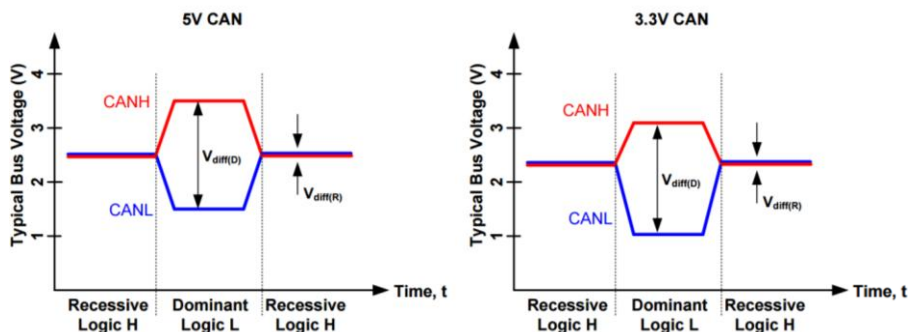


Pin Signal Description:

- 1: CAN_GND = Ground/0 V/V-
- 2: CAN_L = CAN_L bus line (dominant low)
- 3: (CAN_SHLD) = Optional CAN Shield
- 4: CAN_H = CAN_H bus line (dominant high)
- 5: (CAN_V+) = Optional CAN external positive supply

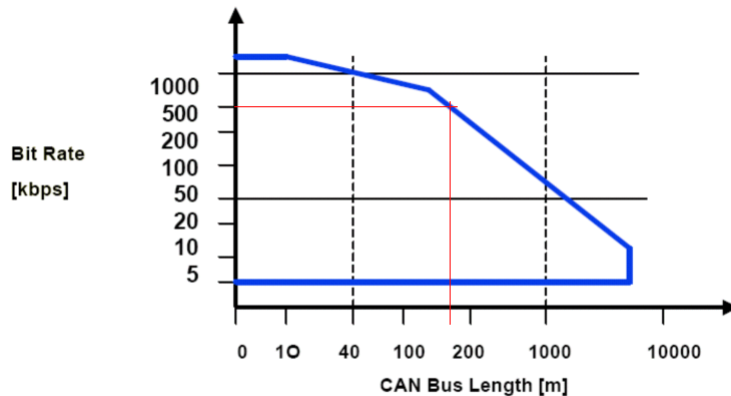
Sinais Elétricos

- Sinais invertidos em tensão e corrente.
 - Melhora imunidade ao ruído
 - Permite utilização degradada com apenas uma linha.



BaudRate vs Bus Length

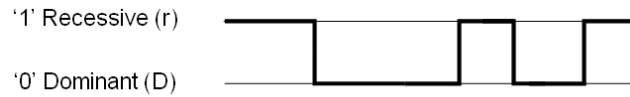
- high-speed (1 Mbits/s) over short distances (40 m)
- low-speed (5 kbits/s) at lengths of up to 10,000 m



Velocidade de transmissão

- Existem diferentes tipos de redes CAN que utilizam velocidades entre 50 e 500 kbit/s.
- Velocidade máxima: 1000 kbit/s.
- Alguns tipos de rede CAN podem funcionar em modo degradado (ex: CAN LS Fault Tolerant).
- Noutras redes (ex: CAN HS), qualquer anomalia provoca a paragem de toda a rede.

CAN Bus Logic



Two logic states on the CAN bus

Node 1	Node 2	Node 3	Bus
D	D	D	D
D	D	r	D
D	r	D	D
D	r	r	D
r	D	D	D
r	D	r	D
r	r	D	D
r	r	r	r

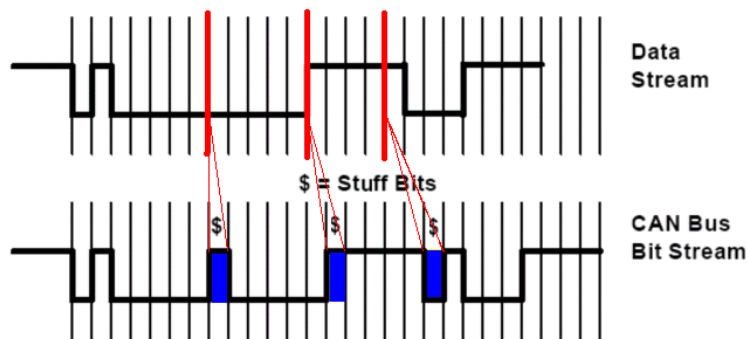
Wired-AND Function

Bus in dominant state

Bus in recessive state
or idle

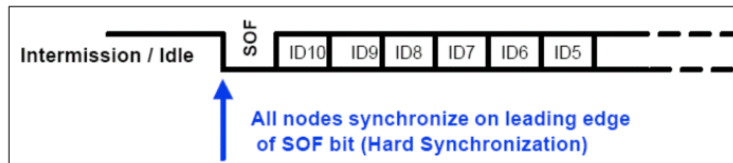
CAN Bit Coding & Bit Stuffing

- NRZ Bit Coding
 - Stuff Bits are inserted after 5 consecutive bits of the same level for synchronization

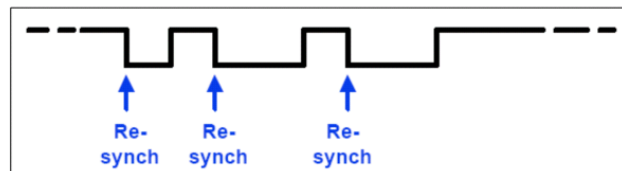


CAN Bus Synchronization

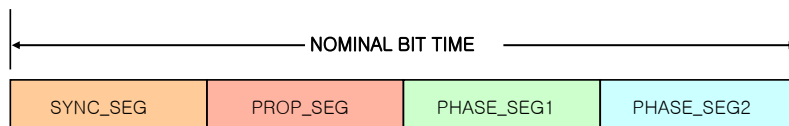
- Hard synchronization at Start Of Frame bit



- Re-Synchronization on each Recessive to Dominant bit



Message Bit Time



SYNC_SEG: Nodes are synchronized within this phase

PROP_SEG: Propagation delay compensation value
[= 2 x (signal propagation time
+ input comparator delay
+ output driver delay)]

Sample Point



PHASE_SEG1 and PHASE_SEG2: Establish correct sampling point

CAN

Data Link Layer

CAN Data Link Layer

- CAN Controller
 - Logical Link Control Sublayer (LLC)
 - Acceptance Filtering
 - Overload Notification
 - Recovery Management

CAN Data Link Layer

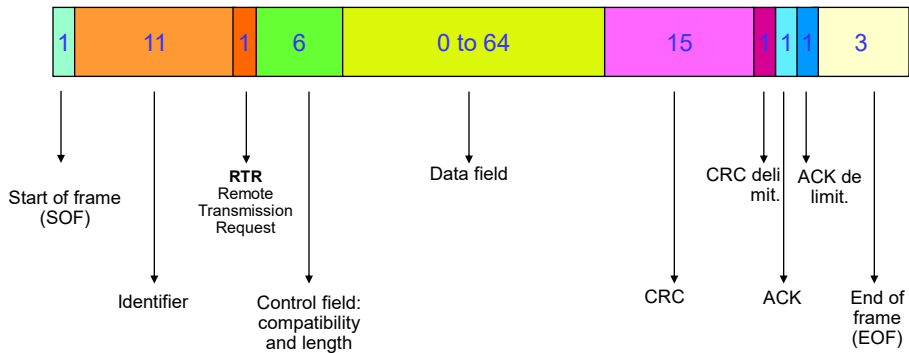
- CAN Controller
 - Medium Access Control Sublayer (MAC)
 - Data Encapsulation/Decapsulation
 - Frame Coding (Stuffing, Destuffing)
 - Medium Access Management
 - Error Detection
 - Error Signaling
 - Acknowledgement
 - Serialization / Deserialization

CAN Frame

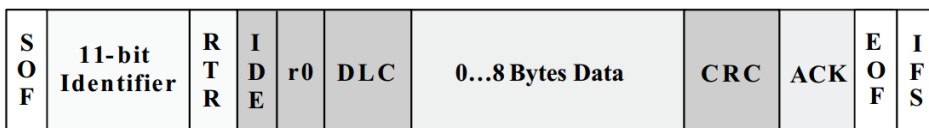
S O F	11-bit Identifier	R T R	I D E	r0	DLC	0...8 Bytes Data	CRC	ACK	E O F	I F S
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- SOF: Start Of Frame
- RTR: Remote Transmission Request
- DLC: Data Length Code
- CRC: Cyclic Redundancy Code
- ACK: Acknowledge
- EOF: End Of Frame
- IFS: Inter Frame Space

CAN Frame

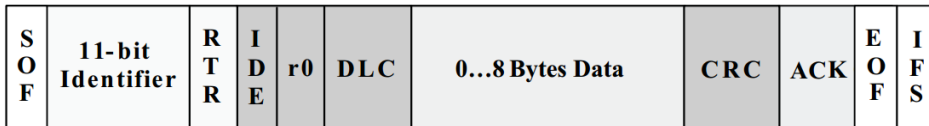


SOF



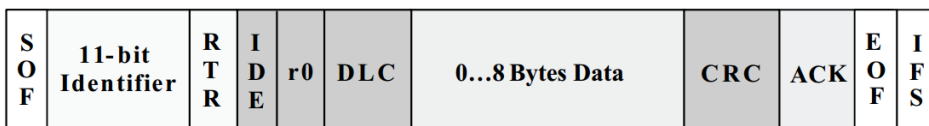
- SOF** – Start Of Frame
 - Single dominant bit
 - Marks the start of a message
 - It is used to synchronize the nodes on a bus after being idle.

Identifier



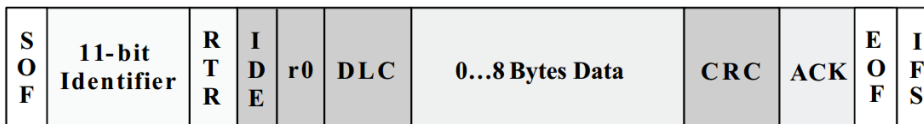
- **Identifier**
 - The Standard CAN identifier is 11 bit long.
 - Máx: 2048 different messages.
 - Identifier establishes the priority of the message.
 - The lower the binary value, the higher its priority.
 - Identifier does not address nodes. It identifies messages.

RTR



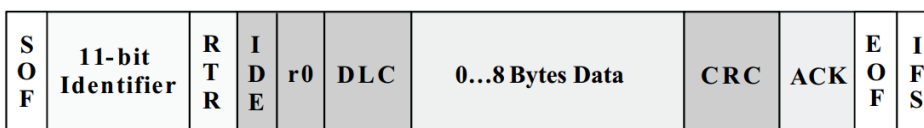
- **RTR** – Remote Transmission Request
 - Single bit.
 - Dominant (0) for data frames.
 - Recessive (1) when information is required from another node.
 - All nodes receive the request, but the identifier is used identify the requested item.
 - The responding data is also received by all nodes and used by any node interested. In this way, all data being used in a system is uniform.

IDE



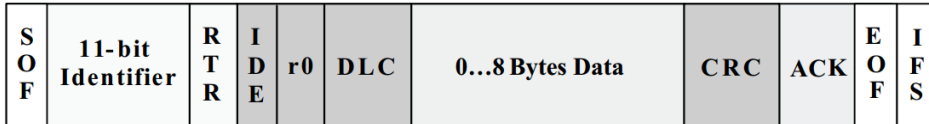
- **IDE** – Identifier Extension
 - Single bit
 - A dominant IDE bit means that a standard CAN identifier with no extension is being transmitted.

r0



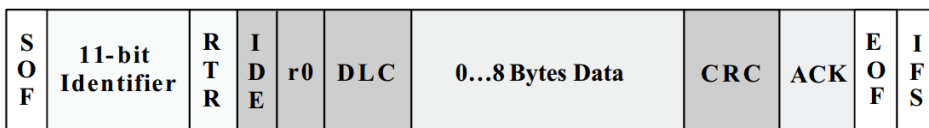
- **r0** – Reserved bit (for possible use by future standard amendment).

DLC



- **DLC** – Data Length Code
 - 4 bit field
 - Contains the number of bytes of data being transmitted.
 - [0-8]

CRC



- **CRC** – cyclic redundancy check
 - 16-bit: 15 bits + 1 bit delimiter
 - Contains the checksum (number of bits transmitted) of the preceding application data for error detection.
 - CRC Delimiter is transmitted as a recessive bit.

ACK

S O F	11-bit Identifier	R T R	I D E	r0	DLC	0...8 Bytes Data	CRC	ACK	E O F	I F S
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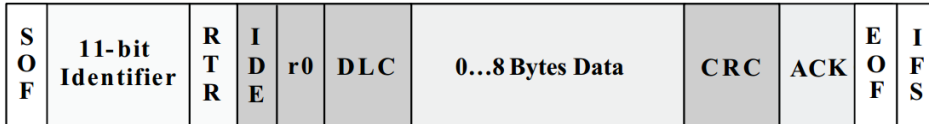
- **ACK – Acknowledge**
 - 2 bits: acknowledgment bit (ACK Slot) + bit delimiter (ACK Del)
- **ACK SLOt**
 - Every node receiving an accurate message overwrites this recessive bit in the original message with a dominate bit, indicating an error-free message has been sent.
 - If message is not acknowledged by any node, it is discarded and the sending node repeats the message.
- **ACK Del** is transmitted as a recessive bit.

EOF

S O F	11-bit Identifier	R T R	I D E	r0	DLC	0...8 Bytes Data	CRC	ACK	E O F	I F S
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- **EOF – End Of Frame**
 - 7 recessive bit field
 - Marks the end of a CAN frame (message) and disables bit-stuffing.
 - Any dominant bit in EOF indicates an error.

IFS



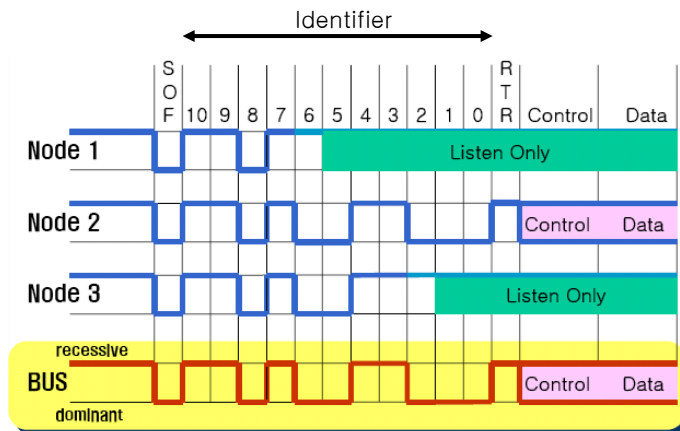
- **IFS** – InterFrame Space
 - 3 recessive bits
 - Contains the time required by the controller to move and process a correctly received frame.

Arbitration

- Each node listens to the data on the network when transmitting.
- When a node transmits a logical 1 but sees a logical 0, it realizes that there is a contention and it quits transmitting.
- Any node that transmits a logical 1 when another node transmits a logical 0 "drops out" or loses the arbitration.
- The node with the lowest identifier has the highest priority.

CAN Bus Arbitration

CSMA/CD with AMP(Arbitration by Message Priority)
=CSMA/BA



CAN Frame Formats

- CAN V2.0A Standard Frame: Identifier is 11 bits long



- CAN V2.0B Extended Frame: Identifier is 29 bits long



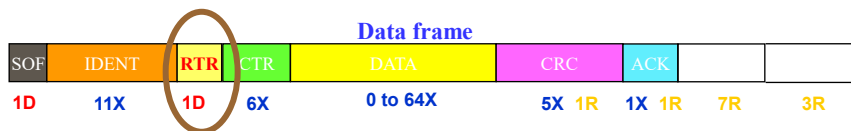
- ID: Identifier
- IDE: Identifier Extension
- RTR: Remote Transmission Request
- SRR: Substitute Remote Request
- RB0/1: Reserved bits
- DLC: Data Length Code

CAN Frame Formats

- Four Frame Types
 - Data Frame (RTR=0)
 - Remote Frame Request (RTR=1 and no data field)
 - Error Frame (after violation frame format)
 - Overload Frame (Error detected during interframe space)

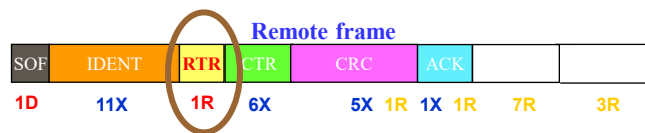
Data Frame

- These frames transport data from a producer to consumers without any guarantee that it will be processed.



Remote Frames

- Remote Frames are used to ask for information.
- Signaled by a recessive (1) RTR bit.
- Sent by a client to a server to request transmission of a data frame.
- The identifier can be shared between remote and data frames.



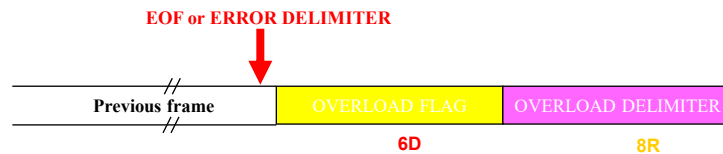
Error Frame

- These frames are transmitted when a station detects the presence of errors on the bus.



Overload Frame

- A node may require an Inter-Frame Space larger than the standard 3 bits.
- An Overload frame is sent if a dominant bit (SOF) is detected during the required IFS.
- These frames are used to ask for an additional time lapse between successive frames (data or request).



Error Detection

- Error Types
 - CRC error
 - Bit-stuffing error
 - Bit error
 - Form Error
 - Acknowledgment error

CAN

Flexible Data Rate

CAN Limitations

CAN Limitations

- Low Data Rate
 - Arbitration imposes: Prop. time $\ll$ Bit time
- Small Message Length
 - 8 Bytes
- ...

CAN FD addresses these 2 limitations:

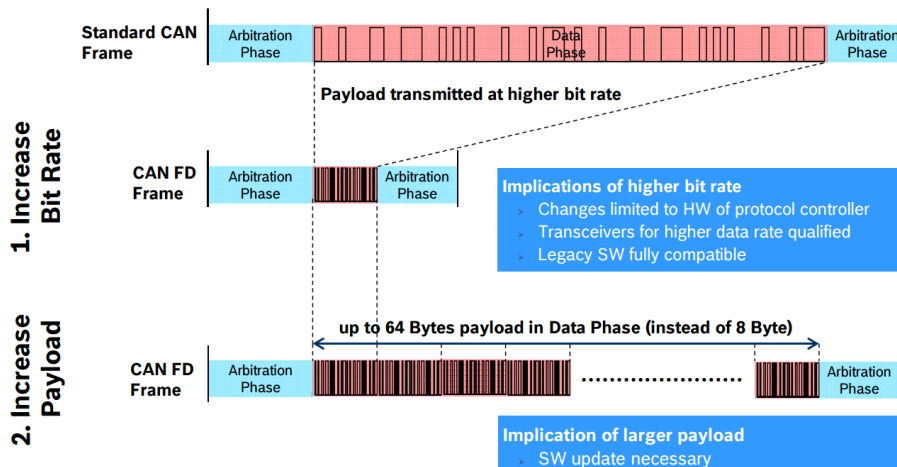
- Improves Data Rate
- Enlarges message length

CAN FD Idea

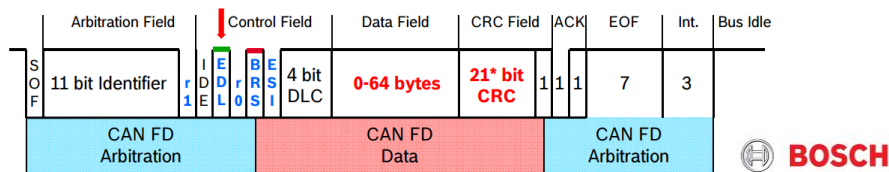
Flexible Data Rate

- Maintain Arbitration bitrate
- Improve Data bitrate
- Allow larger messages

Flexible Data Rate Approach



CAN FD Frame



* 17 bit CRC for data fields with up to 16 bytes

- CAN FD Arbitration Phase
 - length: 30 bit times*
 - data rate: max. 1 MBit/s
- CAN FD Data Phase
 - length: 86 bit times* (8 data bytes)
 - data rate: > 1 MBit/s
- Remote Frames always in CAN Format
 - RTR bit replaced by reserved bit r1
 - r1 takes part in CAN arbitration
 - reserved for protocol expansion

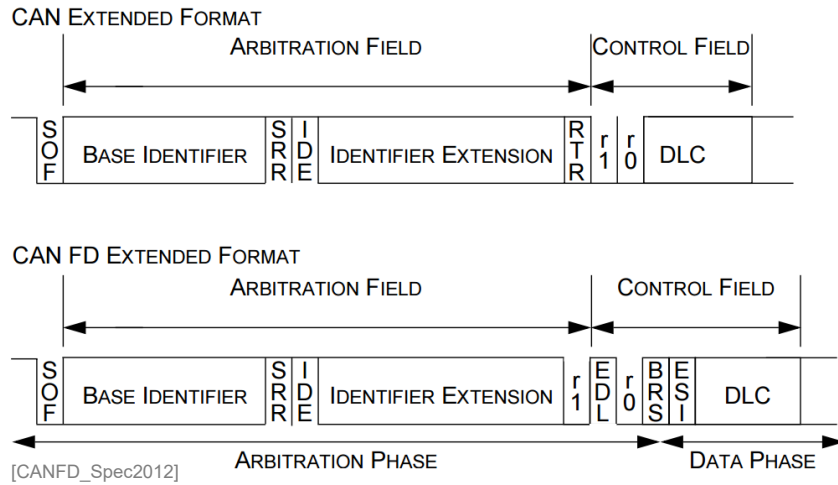
* bit stuffing not considered

ISO 11898-1:2015

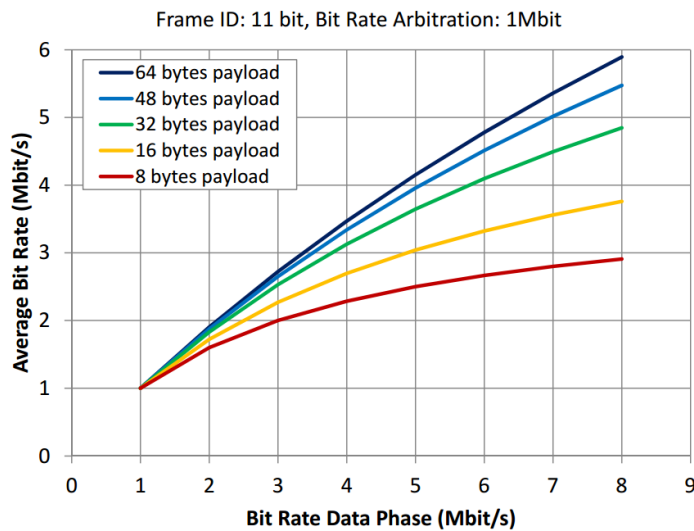
CAN FD Data Length

No. of data bytes	Data length code (DLC)			
	DLC3	DLC2	DLC1	DLC0
0 to 8	As in Classical CAN			
12	r	d	d	r
16	r	d	r	d
20	r	d	r	r
24	r	r	d	d
32	r	r	d	r
48	r	r	r	d
64	r	r	r	r

CAN FD Extended Format



CAN FD Bit Rate



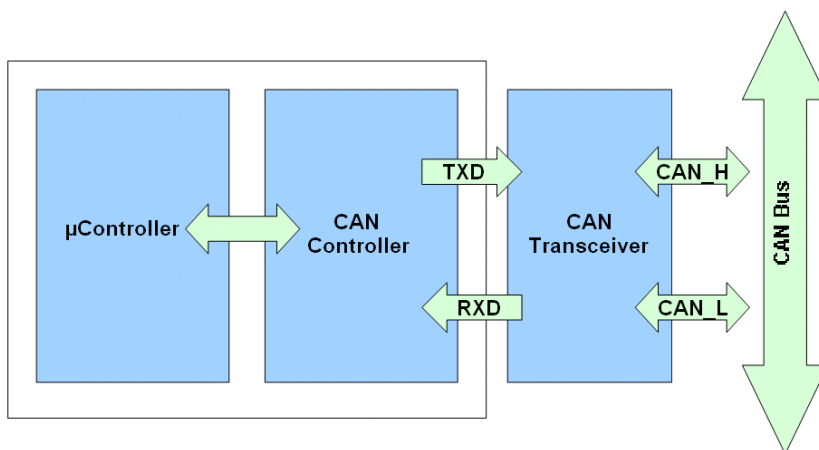
CAN FD Summary

- Upgrade to the classic CAN technology
 - Similar costs as for classic CAN
 - Higher bandwidth
 - Longer messages
 - Small impact on current SW and applications
 - Physical layer and topologies can be maintained
- Life time cycle of existing architectures can be extended.
- High interest by the car industry.

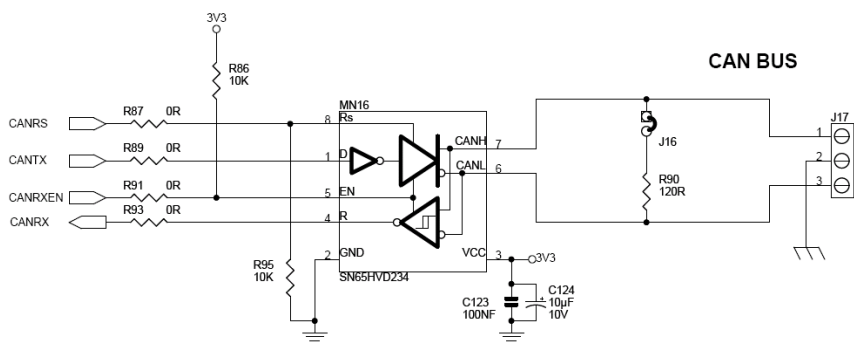
CAN

CAN Controllers

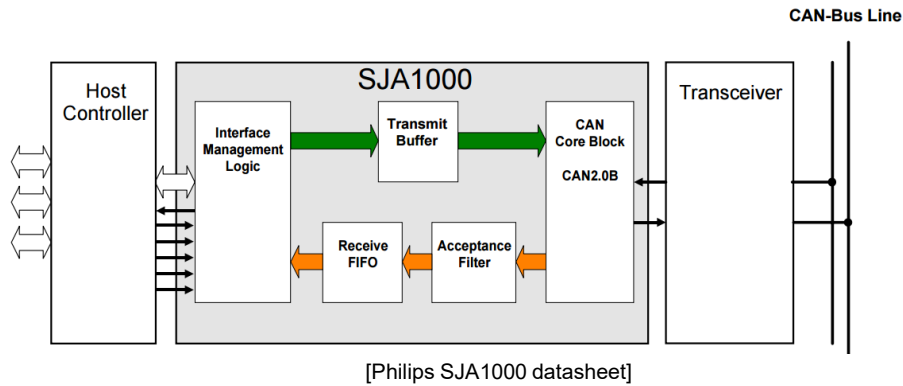
Typical CAN Node



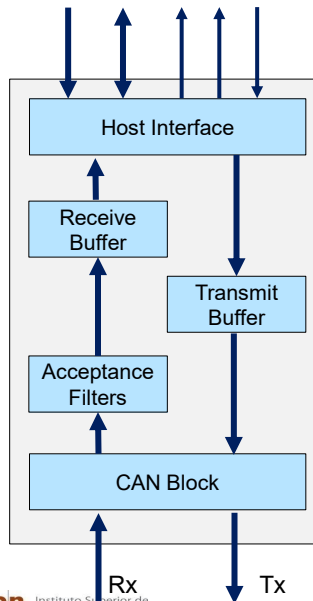
Transceiver



Traditional CAN Controller

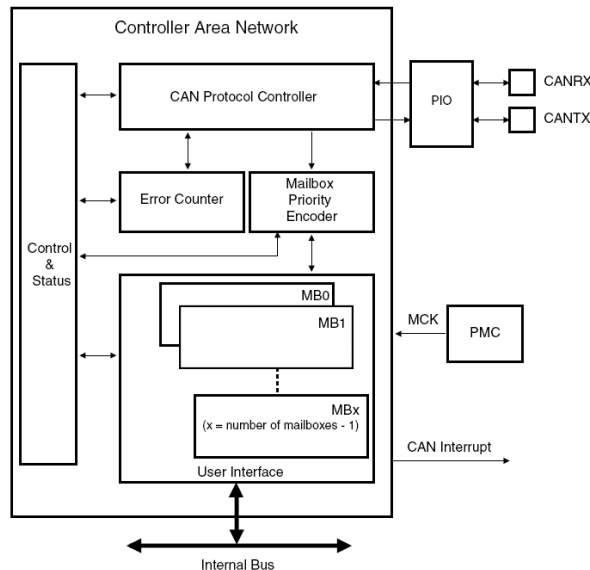


Traditional Controller



Host application Reads and Writes data by transmission order.

CAN Controller with Mailboxes



ATMEL AT91SAM9263 CAN Controller

Mailbox Controller

- Mailboxes
 - Also called channels or buffers.
 - Each mailbox has an ID.
 - Several mailboxes can use the same ID.
 - CPU defines mailbox IDs during initialization.
 - Each mailbox can be configured in receive or transmit mode

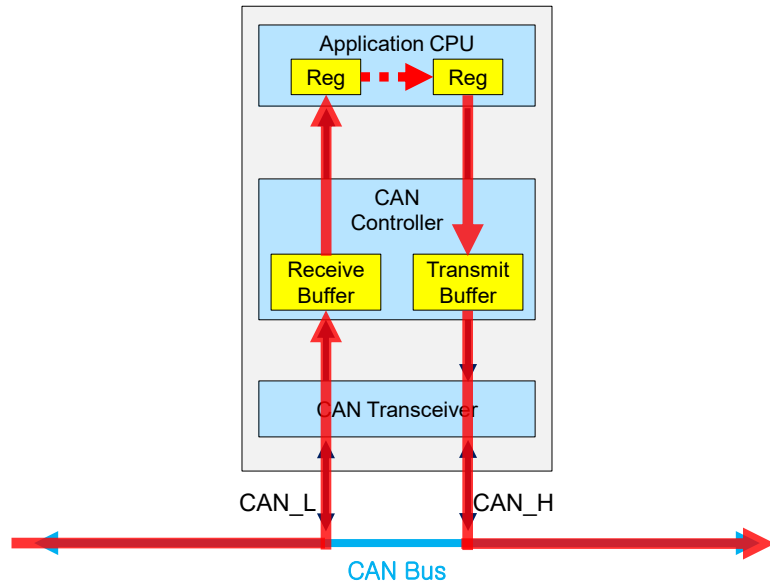
Mailbox Controller

- Reception
 - Data received from data frame is store in a mailbox.
 - Mailbox is selected by comparing message identifier with mailbox ID.
 - The CPU reads data from the CAN controller mailboxes.
 - Data at Mailbox may be overwritten.
 - CPU gets last received data.

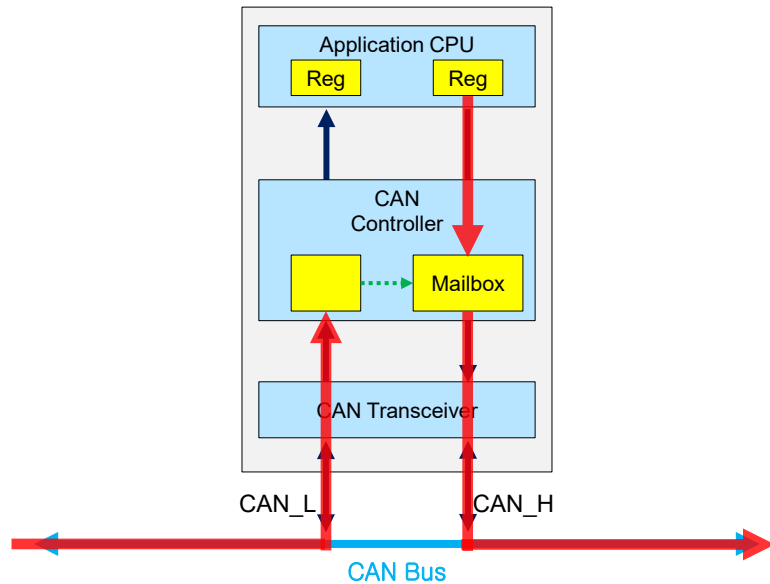
Mailbox Organization

- Sending
 - The CPU writes data to the CAN controller mailboxes.
 - When a Remote Frame Request is received, a Data Frame can be automatically sent.
 - Transmitted data is retrieved from the mailbox selected based on message ID.
 - Application on CPU is not involved
 - Latency is reduced

Traditional Operation



Mailbox Operation



AT91SAM9263 CAN Controller

- CAN 2.0A and CAN 2.0B
- Bit rates up to 1Mbit/s
- 16 Mailboxes
 - Each can be Receive or Transmit
 - Local Tag and Mask Filters
 - 16-bit Time Stamp
 - 32x2 Data Register

Mailbox Priority

- Reception Mode
 - Mailbox with the lowest number is serviced first.
- Transmission Mode
 - Register CAN_MMR (Mailbox Mode Register)
 - PRIOR:4bit – mailbox priority
 - Same priority -> mailbox with lowest number first